

Research Proposal (Selected Objective)

Research direction: Studying lightweight, fairness-preserving, and privacy-protecting AI-based personalized learning methods for low-resource educational scenarios

English translation of the Chinese proposal; the Chinese version and its review checklist remain authoritative.

I. Needs analysis

In the field of education, artificial intelligence personalized learning technologies aim to address diverse student needs through adaptive content delivery and real-time feedback, yet significant challenges persist in practical applications. Particularly in low-resource settings, student groups struggle to obtain targeted support due to individual learning differences, while teachers endure high teaching burdens and skill gaps. Relevant studies indicate that 83% of schools in rural China lack essential hardware equipment, limiting the deployment of AI tools; concurrently, teacher readiness requires 200+ hours of training, exacerbating teacher shortages. Schools and administrative departments need to address uneven resource allocation, such as urban-rural infrastructure disparities hindering the widespread adoption of personalized learning, thereby increasing the burden on educational equity. These demands stem from real-world resource constraints and capacity deficiencies, not theoretical assumptions.

Existing systems struggle to adequately meet these requirements, primarily constrained by three categories of obstacles: cost and scalability, fairness, and privacy and compliance. First, high development costs and computational resource demands impede large-scale implementation. For instance, certain methods rely on GPU clusters, while inadequate device coverage in low-resource environments restricts deployment scope. Second, fairness issues are prominent, as algorithms may amplify demographic disparities—for example, prediction outcomes influencing subsequent data through feedback loops, exacerbating inter-group inequalities—and existing technologies insufficiently address socio-emotional factors. Finally, privacy protection technologies face bottlenecks, with performance significantly degrading in low-bandwidth environments, while differing domestic and international compliance requirements increase implementation complexity. These intertwined obstacles make it difficult for current solutions to simultaneously achieve lightweight design, fairness, and privacy protection.

This project focuses on low-resource educational scenarios in China, particularly rural schools and online education platforms, which are constrained by weak infrastructure and resource scarcity. Relevant research shows a 40% hardware deficiency rate in rural areas, and technologies like Federated Learning experience latency surges of 3.7 times in low-bandwidth networks, highlighting the specificity of these scenarios. Such environments require solutions that must adapt to limited computational resources, high student-teacher ratios, and domestic privacy regulations (e.g., PIPL (Personal Information Protection Law)), avoiding the introduction of cross-border elements.

Centered on the theme of "achieving lightweight, fair, and privacy-preserving personalized learning under low-resource constraints," the core research questions include: First, how to design a lightweight learning architecture that significantly reduces computational resource dependence while maintaining core task performance; Second, based on this architecture, how to optimize fairness mechanisms to prevent bias amplification in data feedback loops; Third, how to integrate privacy protection technologies to ensure efficient operation in low-bandwidth environments. These questions progress sequentially from foundational resource optimization to fairness enhancement, ultimately achieving privacy compliance, aligning with the direction of

research needs.

II. Research status

In the field of personalized learning modeling and prediction, machine learning methods construct predictive models using students' historical data. Techniques such as random forests and support vector machines have been applied to career recommendation (e.g., achieving 93% accuracy in a systematic review, 2025). However, such models rely on high-quality quantitative data, exhibit insufficient adaptability to sparse data in low-resource scenarios, and may amplify group biases (e.g., studies show African American students were mislabeled as high-risk at 2.3 times the probability of white students).

Intelligent tutoring and generative feedback technologies focus on real-time interactive support. The current mainstream trend shifts toward generative AI. Its limitations include a 15-20% hallucination rate in open scenarios as indicated by relevant research, and teachers requiring over 200 hours of training for effective deployment, with significant real-time feedback latency in low-bandwidth environments.

Privacy-preserving technologies ensure data security through encryption and distributed computing. Federated learning combined with differential privacy meets compliance requirements (e.g., GDPR and PIPL). However, according to related studies, federated learning experiences a 3.7-fold increase in latency under low-bandwidth networks, and compliance discrepancies exacerbate cross-regional deployment complexity.

Fairness assurance mechanisms explore algorithmic debiasing and synthetic data. Generative Adversarial Networks (GANs) generate synthetic data for learners with special needs, reducing group bias by 32% in Indian literacy applications. Nevertheless, synthetic data carries risks of mode collapse, and ethical controversies remain unresolved. Current research predominantly relies on centralized computing power (83% of rural schools lack GPU clusters), making it difficult to meet lightweight requirements in low-resource scenarios.

In summary, existing technologies individually optimize single metrics, lacking synergistic design for low-resource scenarios: lightweight models often sacrifice fairness, privacy protection intensifies computational demands, and synthetic data relies on high-cost infrastructure. There is a need to establish an end-to-end framework that balances computational efficiency, group fairness, and privacy protection, particularly addressing deployment feasibility under hardware deficiencies in rural schools and low-bandwidth constraints.

III. Research objective

This study aims to develop a resource-efficient AI framework for deployment on low-resource educational devices (such as those commonly found in rural schools), addressing critical issues in computational complexity, algorithmic bias, and privacy latency inherent in existing methods. Specifically, the high computational complexity of existing machine learning and reinforcement learning methods (requiring GPU cluster support) prevents deployment in low-resource environments like 83% of rural schools in China, violating resource-efficiency requirements; algorithmic bias in AI-assisted assessment tools and machine learning methods amplifies demographic disparities, such as a risk prediction bias ratio reaching 2.3 times, undermining fairness guarantees; privacy-enhancing technologies experience latency spikes on low-bandwidth networks, conflicting with resource-efficient deployment, and integrated solutions combining resource efficiency, fairness, and privacy protection are lacking. The objective is, through innovative design, to reduce computational requirements by 50%, decrease

the fairness disparity ratio by 40% (specifically manifested as reducing the risk prediction bias ratio from 2.3 times to an expected ≤ 1.5 times), and control privacy protection latency within 100 milliseconds, thereby providing a personalized AI learning method framework for low-resource educational settings that balances resource efficiency, fairness, and privacy protection.

IV. Research plan

1. Research content

This study will conduct systematic work on a unified benchmark model using the same primary dataset. First, a neural network-based personalized learning benchmark model will be constructed for student ability mapping and adaptive path optimization. Baseline metrics including accuracy, R^2 , and risk prediction bias ratio will be measured on Kaggle's "AI-Driven Personalized Learning Dataset". Subsequently, innovative mechanisms will be integrated layer by layer:

The lightweight layer employs INT8 quantization to transform model weight precision, addressing the high computational demands of existing machine learning methods for deployment on low-resource devices;

The fairness layer embeds adversarial loss functions during training to dynamically optimize algorithm biases related to demographic variables;

The privacy-preserving layer injects Laplace noise during gradient updates to achieve privacy protection under low-bandwidth network constraints.

The core innovation lies in analyzing interactions among the three-layer mechanisms—quantization compression may amplify biases against specific groups, privacy noise may interfere with fairness optimization, while computational resources saved through lightweight design can partially offset additional latency overhead introduced by these mechanisms. This multi-objective collaborative optimization mechanism is key to overcoming current technological silos.

2. Approach and methods

A full factorial experimental design is employed to validate mechanism efficacy. Building upon the baseline model, eight sets of comparative experiments are sequentially executed: Group 1 disables all mechanisms as the baseline; Groups 2 to 4 individually enable the lightweighting, fairness, or privacy-preserving mechanism; Groups 5 to 8 enable dual-mechanism or full-mechanism combinations. Each group of experiments measures a unified set of metrics on the same dataset: model performance (accuracy, R^2 , MSE), computational efficiency (CPU inference latency), fairness (risk prediction bias ratio, equality of opportunity difference, demographic parity difference), and privacy overhead (additional latency overhead relative to the baseline after mechanism introduction, (ϵ, δ) -differential privacy parameters, sample-level granularity). Main effects (e.g., the impact of solely enabling lightweighting on latency) and interaction effects (e.g., whether the latency change when lightweighting and privacy-preserving are jointly enabled deviates from the sum of their independent effects) are quantified by comparing inter-group differences. Target values, such as the risk prediction bias ratio being reduced to the expected level relative to the baseline, latency controlled within 100ms under specified simulated device environments, and model accuracy degradation not exceeding 5%, must all be empirically derived through the aforementioned inter-group comparisons.

3. Comparative advantages

In the dimension of lightweighting, INT8 quantization offers superior hardware compatibility compared to knowledge distillation and better preserves structural integrity than model pruning, meeting deployment constraints on GPU-less devices. For fairness mechanisms, adversarial debiasing is selected because reweighting exhibits limited effectiveness for sparse groups, synthetic data generation may introduce new biases, while adversarial training can directly optimize the correlation with sensitive attributes. For privacy protection, differential privacy is adopted instead of homomorphic encryption or secure aggregation, as it maintains controllable latency in low-bandwidth environments and requires no additional communication protocols. All technical approaches are based on mature solutions: INT8 quantization belongs to the standard machine learning toolchain, adversarial training is widely applied in the field of AI fairness, and differential privacy possesses theoretical guarantees within privacy-enhancing technologies. The selected solutions collectively support the core innovation path of "integrated three-layer optimization within a single model".

4. Feasibility

At the theoretical level, INT8 quantization reduces memory access overhead by lowering numerical precision, adversarial debiasing suppresses sensitive attribute correlations through gradient reversal, and differential privacy provides provable guarantees via noise injection—all three mechanisms have been academically validated. Technologically, the baseline model is constructed using the PyTorch framework, lightweighting is achieved through `torch.quantization`, the fairness module integrates the Fairlearn adversarial debiasing component, and privacy protection is implemented based on the Opacus differential privacy library, forming a comprehensive technical stack. Regarding experimental conditions, publicly available educational datasets are employed for validation, with the experimental environment simulating low-resource devices by containerizing CPU cores, memory, and network bandwidth. Data quality undergoes three-step verification: authenticating dataset validity, checking completeness of demographic fields, and evaluating group distribution balance. Should essential attributes be missing from the primary data, the OpenML educational fairness benchmark datasets will be utilized for supplementary validation.